

Generalised Bidirectional ODN Characterisation with Unknown Number of Loss Events for Coherent PONs

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Abstract We propose a generalised bidirectional DSP-based framework using adaptive Gaussian process regression and Bayesian information criterion for precise and efficient ODN characterisation in coherent PONs with unknown loss events. Experiments demonstrate accurate dual-loss localisation and quantification, with maximum mean errors of 0.26km and 0.17dB. ©2026 The Author(s)

Introduction

Coherent passive optical networks (PONs) are emerging as a promising solution for next-generation access systems due to their high sensitivity, extended reach, and advanced digital signal processing (DSP) capability [1, 2]. Beyond data transmission, coherent transceivers also provide a valuable platform for network monitoring [3, 4].

Recently, DSP-based ODN characterisation has emerged that exploits deployed coherent transceivers for physical-layer sensing without extra monitoring hardware [5, 6]. However, characterising high-loss ODNs remains challenging, since large insertion losses introduced by high-ratio splitters weaken the physical signatures for monitoring. Our previous work [6] showed that bidirectional ODN characterisation is feasible under single high-splitting loss cases using forward-propagation and Gaussian process regression (GPR) [7].

In practice, however, modern PONs may contain more than one high-loss event, for example due to cascaded splitters, local defects, or poor connections. Moreover, the number of loss events is not known a priori. Therefore, a more general framework is required to move beyond the single-loss assumption, enabling ODN characterisation while also performing model order selection.

In this paper, we propose a generalised bidirectional framework for ODN characterisation in coherent PONs. Our method combines an adaptive GPR surrogate model with Bayesian information

criterion (BIC)-based model order selection. This eliminates the requirement for a priori knowledge regarding the number of loss events and enables joint estimation of the fibre attenuation coefficient alongside the locations and magnitudes of discrete losses. Our approach is validated by simulation and experimental results, achieving accurate ODN characterisation in dual-loss cases.

Principle and Model Design

The proposed method formulates bidirectional ODN characterisation with unknown number of loss events as a constrained inverse parameter estimation problem, shown in Fig. 1. In the physical path, transmitted signals undergo real ODN link and DSP to obtain measured post-DSP signals in both downstream (DS) and upstream (US) directions. In the forward-propagation evaluation block, the corresponding received signals for each candidate ODN model are then simulated by solving the Manakov equation using the split-step Fourier method (SSFM) and processed by the same DSP chain. The signal power at distance z in an ODN of length L is modelled by distributed fibre attenuation together with k discrete high-loss events as

$$P_{\text{dB}}(z) = P_0 - \alpha z - \sum_{i=1}^k S_i H(z - z_i), \quad (1)$$

where P_0 is the launch power, α is the fibre attenuation coefficient, and z_i and S_i denote the position and magnitude of the i^{th} lumped loss, respectively.

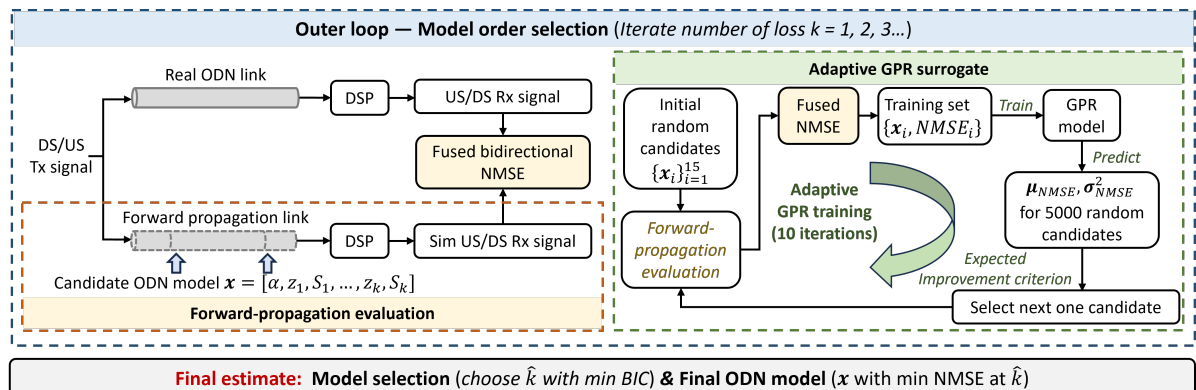


Fig. 1: Block diagram of the proposed generalised bidirectional ODN characterisation.

For a fixed number of loss k , the unknown parameter vector is $\mathbf{x} = [\alpha, z_1, S_1, \dots, z_k, S_k]$.

For each candidate ODN model with parameter vector $\mathbf{x}$, the measured and forward-propagated DS and US post-DSP signals are compared, and the mismatch in each direction is quantified by the normalised mean squared error (NMSE) [6] as

$$\text{NMSE}_d(\mathbf{x}) = \frac{\sum_{n=1}^N \|\mathbf{R}_d[n] - \hat{\mathbf{R}}_d[n; \mathbf{x}]\|^2}{\sum_{n=1}^N \|\mathbf{R}_d[n]\|^2}, \quad (2)$$

with N the number of samples. $\mathbf{R}_d[n]$ and $\hat{\mathbf{R}}_d[n; \mathbf{x}]$ are the measured and simulated post-DSP samples, and $d \in \{\text{DS}, \text{US}\}$. Fused bidirectional NMSE is obtained by averaging DS and US NMSEs, ensuring equal contribution from both directions. Compared with single-direction characterisation, the bidirectional formulation provides stronger constraints on the inverse problem, reducing solution ambiguity and improving search efficiency.

As the model order k increases, a direct exhaustive search over the large parameter space becomes impractical since each objective evaluation requires expensive bidirectional forward-propagation and DSP. Therefore, an adaptive GPR surrogate model [8] is employed to provide a computationally efficient search process. To keep the search physically reasonable, the sum of the fibre attenuation and discrete losses is constrained to match the measured end-to-end loss.

The framework of the adaptive GPR surrogate is illustrated in the left panel of Fig. 1. For each value of k , an initial small set of 15 candidates $\{\mathbf{x}_i\}_{i=1}^{15}$ is randomly generated and evaluated through forward-propagation evaluation to obtain the fused NMSE values. The surrogate GPR model is then trained on this initial training set using a radial basis function (RBF) kernel [7]. Next, the trained GPR model predicts the mean μ_{NMSE} and uncertainty σ_{NMSE}^2 for a pool of 5,000 randomly sampled ODN candidates. Based on these predictions, the Expected Improvement criterion [9] is used to adaptively select the next candidate, prioritising values that are either promising or uncertain. This ensures that the expensive SSFM-based forward-propagation evaluation is performed only for single most informative candidate. The resulting fused

NMSE value and the selected candidate are appended to the training set, after which the GPR model is retrained to select the next candidate from a newly generated pool of 5,000 samples. Owing to its fast convergence, the adaptive training loop is limited to 10 iterations.

The number of loss events k is determined in an outer loop using the BIC [10, 11] defined as:

$$\text{BIC}(k) = n \ln(\text{NMSE}_{\min}(k)) + p_k \ln n, \quad (3)$$

where $\text{NMSE}_{\min}(k)$ is the minimum fused NMSE in k -loss ODN model, denoting the best model for order k and serving as the model fitting error in the BIC formulation, p_k is the number of model parameters, and n is the number of measured received samples. The final model order is chosen by minimising the BIC. The corresponding parameter estimate with the minimum NMSE serves as the final characterisation result. Thus, the inner adaptive GPR surrogate search (for a fixed model order) and the outer BIC-based model order selection jointly determine the final ODN link estimate.

Simulation Results

The proposed model was first evaluated in simulation. The transmission link consisted of a 40 km fibre span with parameters of $\alpha = 0.2$ dB/km, $\beta_2 = -21.6$ ps²/km, $\gamma = 1.32$ W⁻¹/km⁻¹. A 9 dB and a 6 dB loss were inserted at 13 km and 32 km in the downstream direction, respectively. A dual-polarisation (DP) 16QAM 60 GBd signal ($N = 65536$ samples) was transmitted at a launch power of 10.5 dBm with a 150 kHz laser linewidth. At the receiver, the standard DSP was applied to obtain the post-DSP samples.

Fig. 2(a) shows the fused bidirectional NMSE versus the number of total forward-propagation evaluations, comparing with and without adaptive training. We focus on forward-propagation counts, as they dominate the computational cost. For the dual-loss case, one GPR training and prediction step takes about 0.014 s, whereas a single bidirectional forward-propagation with DSP requires about 1.45 s, over 100× more computation. By replacing expensive forward-propagation evaluations with fast surrogate-based searches, adaptive GPR enables efficient parameter space exploration and achieves lower NMSE at the same com-

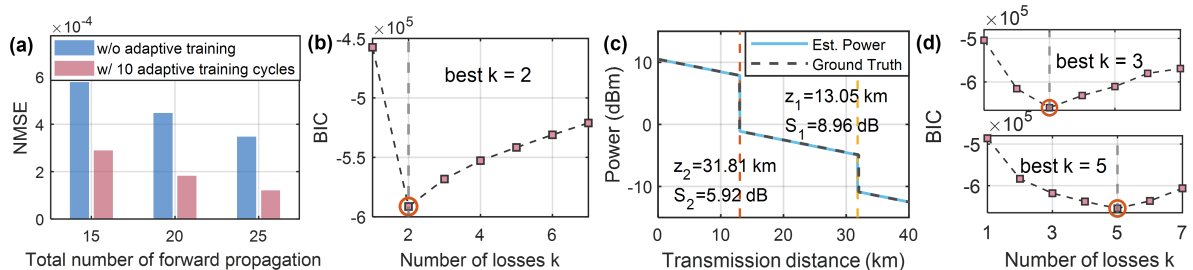


Fig. 2: Simulation-based ODN characterisation results. (a) Fused NMSE vs. total number of forward propagations. (b) Model order selection with BIC value vs. number of losses. (c) Estimated power profile with loss location and magnitude estimation. (d) Model order selection using BIC with 3×5 dB loss at 10, 20, 30 km (upper panel) and 5×3 dB loss at 8, 16, 24, 32, 40 km (lower panel).

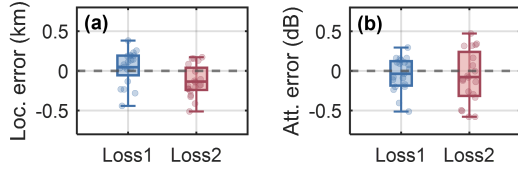


Fig. 3: Simulation results of (a) localisation and (b) quantification error distribution of the estimation of two losses.

putational cost. The NMSE improvement saturates beyond approximately 25 evaluations. Therefore, we selected 15 initial candidates followed by 10 adaptive iterations in our implementation.

Fig. 2(b) shows the BIC reaching its minimum at $k = 2$, which correctly matches the simulation settings. After selecting the optimal model order, the longitudinal power profile was reconstructed based on the best candidate ODN model at this specific model order, with the result representing the mean over 20 independent characterisation, as shown in Fig. 2(c). Results demonstrate good alignment between the estimated power profile and the ground truth, with maximum mean error of 0.19 km and 0.08 dB, respectively. The fibre attenuation coefficient was estimated to be 0.203 dB/km.

While this dual-loss case is presented for illustration, the proposed framework is more general and not limited to two-loss scenarios. The BIC values for the three-loss and five-loss cases is shown in Fig. 2(d). To maintain a total loss of 15 dB across all cases, consistent with the dual-loss scenario, either three 5 dB losses or five 3 dB losses were uniformly distributed over the range [0,40] km. The minimum BIC in both cases clearly confirm the proposed framework extends to higher-order loss scenarios.

To further demonstrate the robustness and precision of the method, the localisation and quantification error distribution of 20 independent characterisation of dual-loss case were shown by boxplots with overlaid scatter points in Fig. 3. The boxplots indicate the interquartile range, the central line denotes the mean, and the whiskers show the extremes. Results indicate the location and magnitude error of two losses remains predominantly below 0.2 km and 0.3 dB, which validates the accurate ODN characterisation achieved.

Experimental Results

The proposed framework was further validated experimentally with the setup shown in Fig. 4(a)

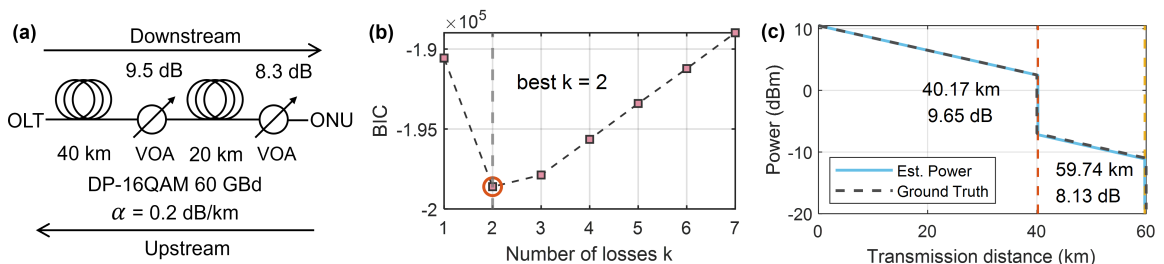


Fig. 4: Experiments for the proposed ODN characterisation. (a) Experimental setup and parameters. (b) BIC value vs. number of losses, validating $k = 2$ as the best model. (c) Measured and estimated power profile with loss location and magnitude estimation.

and follows our previous coherent PON setup in [6]. The 60 GBd DP-16QAM signals were transmitted in a 60 km fibre link for both US and DS. A 9.5 dB and a 8.3 dB high loss were inserted at 40 km and 60 km in the downstream direction to simulate the high splitting loss events. As in simulation, the same framework from Fig. 1 was implemented.

Fig. 4(b) shows that the minimum BIC is obtained at $k = 2$, which is consistent with the true experimental configuration and confirms that the proposed outer loop model order selection can correctly identify the number of dominant discrete losses. Fig. 4(c) shows the estimated longitudinal power profile, which closely matches the true attenuation trend and the two dominant loss events compared with the ground truth. The maximum mean localisation and quantification errors for two losses were 0.26 km and 0.17 dB, respectively.

The localisation and quantification error distribution are shown in Fig. 5(a) and (b), respectively. For Loss 2, the distributions exhibit a slight negative bias since the second loss is located at the fibre end, where the localisation error becomes one-sided. Overall, the errors remain within a limited range, confirming that the proposed method provides stable estimation in experiments.

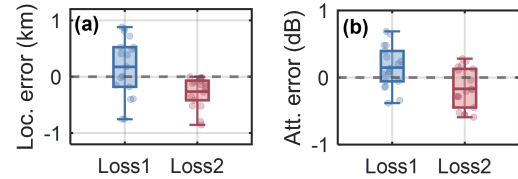


Fig. 5: Experiment results of (a) localisation and (b) quantification error distribution of the estimation of two losses.

Conclusions

We proposed a bidirectional DSP-based framework for ODN characterisation in coherent PONs with an unknown number of high-splitting loss events using an adaptive GPR surrogate model and Bayesian information criterion. Using bidirectional post-DSP signals, the framework jointly estimates the number of loss events and reconstructs the longitudinal power profile, including fibre attenuation and discrete loss localisation and quantification. Simulation demonstrate the generality of the approach. Moreover, experimental results achieve a maximum mean localisation and quantification errors of 0.26 km and 0.17 dB, respectively.

Acknowledgements

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